AMERICAN TERAWATT

HVDC System Functional Specification

Private Industrial Power Grid

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Classification	Confidential

Revision history

Rev.	Date	Author	Description
0.1	August 16, 2026	E. Starschich	Initial draft
0.2	August 17, 2026	E. Starschich	Implementing first review comments

Contents

1 Introduction

1.1 Purpose

This functional specification defines the functional and performance requirements for the HVDC transmission system to be supplied for the *Utah/Nevada Industrial Grid (UNIG)* scheme and is the basis for HVDC system supplier selection.

American Terawatt is developing the Utah/Nevada Industrial Grid (UNIG), a private industrial grid that connects clean firm generation in Utah, Nevada and the wider region directly to hyper-scale data-center load. Discussions with generators are ongoing to unlock 8–10 GW+ of clean firm power for this demand. Transmission buildout has not kept pace with demand growth in Nevada and Utah, leaving clean generation without a path to market; UNIG is intended to close that gap outside the conventional interconnection process.

UNIG is built in stages. The first stage comprises four point-to-point HVDC links based on a standardized HVDC design. In the second stage these links are networked by additional station-to-station links to form a meshed system, giving higher availability and greater flexibility in routing power from generation to load. The resulting conceptual network is shown in Figure 1.

The second stage should not have an impact on the time schedule of the Phase 1 development and it is understood that terminal modifications from Phase 1 are necessary to allow the Phase 2 configuration.

The project is executed on an accelerated schedule. The 30 % design and routing studies are underway and are expected to be completed by the end of November 2026. The requirements, in particular the standardized hardware design and the staged buildout capability, follow directly from the schedule given in Section 7.

1.2 Scope

The scope of supply is the HVDC terminals, together with any additional equipment that the supplier considers necessary for the integration of the HVDC system into the grid.

1.3 Key project characteristics and design drivers

The following characteristics drive the design and execution of the scheme and should be considered by the supplier:

- **Islanded operation.** The system operates as an islanded network with *no interconnection to the wider AC grid*. The HVDC link and its connected generation and loads form an islanded network, which the converter control must be able to establish and stabilise.
- **Staged generation and transmission buildout.** Generation is built out in 300 MW steps, with the pace set by water availability. staged buildout of the transmission line, phased to match the generation ramp, is therefore beneficial and is a design objective.
- **High reliability requirements.** The data-center loads impose high reliability and availability requirements on the transmission system (see Section 5.1).
- **Standardized hardware design.** A standardized hardware design approach shall be adopted to reduce hardware lead times and to limit configuration variations between schemes.

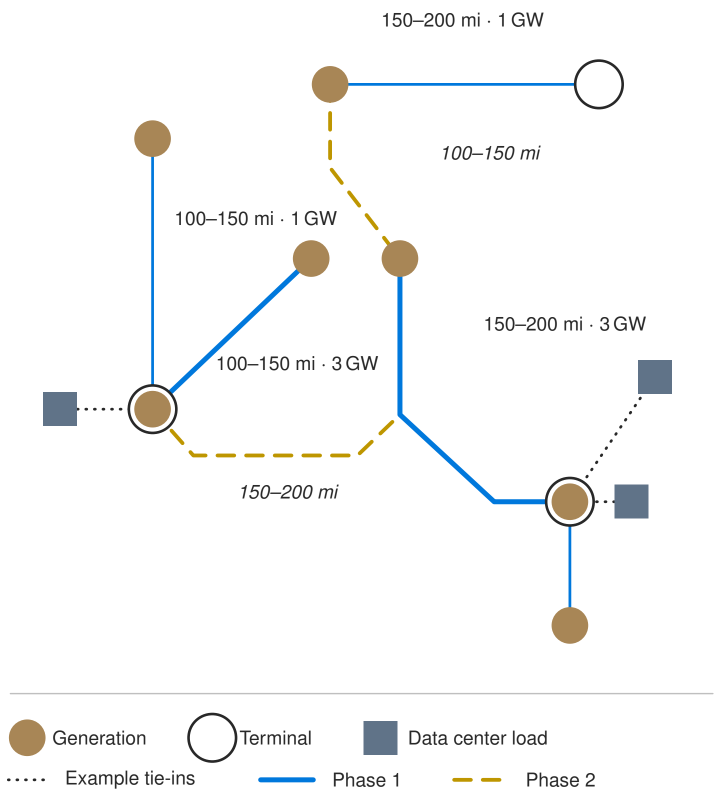

Figure 1: UNIG conceptual diagram.

2 System overview

2.1 Grid configuration

The HVDC system is a point-to-point, bidirectional VSC-HVDC link connecting the converter stations via a DC transmission line. An overview of the system with HVDC scope of supply is shown in Figure 2.

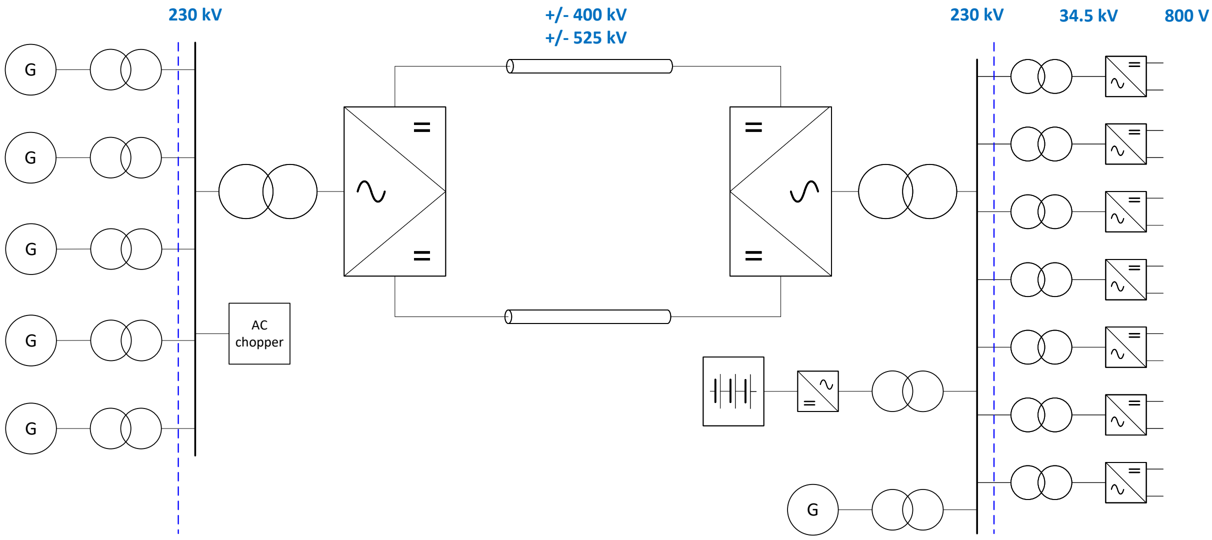

Figure 2: System overview with HVDC system scope of supply

2.2 Transmission schemes

The transmission system will utilize two HVDC schemes. The schemes differ in their power rating, which might result in different transmission topologies. An overview of the technical parameters is given in Table 1.

Parameter	Description	Scheme 1	Scheme 2
P_{AC_Rec}	Rated active power (AC rectifier)	1000 MW	3000 MW
Q_{AC_Rec}	Rated active power (AC rectifier)	±330 MVAr	± 990 MVAr
U_{DC}	Rated DC voltage (pole–ground)	±400 kV	±400 kV, ±525 kV
Config.	Transmission configuration	SMP	DSMP or BIP with DMR
L	DC line length	~200 mi	~200 mi
f	AC system frequency	60 Hz	60 Hz

Table 1: HVDC configuration: Scheme 1 and Scheme 2.

For the reactive power an initial power factor of $\cos\phi = 0.95$ was defined. As the system investigations are ongoing, it is expected that the reactive power requirements can be significantly reduced.

2.3 HVDC topologies

2.3.1 Scheme 1

The anticipated transmission configuration for Scheme 1 is a symmetrical monopole with a DC voltage rating of ±400 kV (SMP, Figure 3). The 400 kV is considered to be the most economical solution for the transmission line considering converter and DC line costs. HVDC supplier is encouraged to alternative solutions in case of significant savings with smaller DC voltage set-up such as ±320 kV.

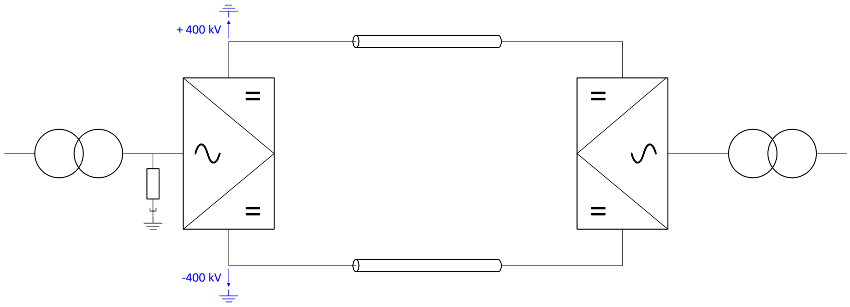

Figure 3: Symmetrical monopole (SMP) configuration.

2.3.2 Scheme 2

For the 3000 MW Scheme 2, two transmission configurations are considered:

1. Dual symmetrical monopole (DSMP, Figure 4) with ±400 kV DC voltage
2. Bipole with dedicated metallic return (BIP, Figure 5) with ±525 kV DC voltage

The DSMP configuration is assumed to provide higher availability and greater flexibility for the staged buildout of the transmission line. Furthermore, the same converter configuration can be used for Scheme 1, taking into account an overrating of the Scheme 1 configuration.
The decision between the DSMP and BIP configurations will be based on the price difference in the converter and DC line costs.

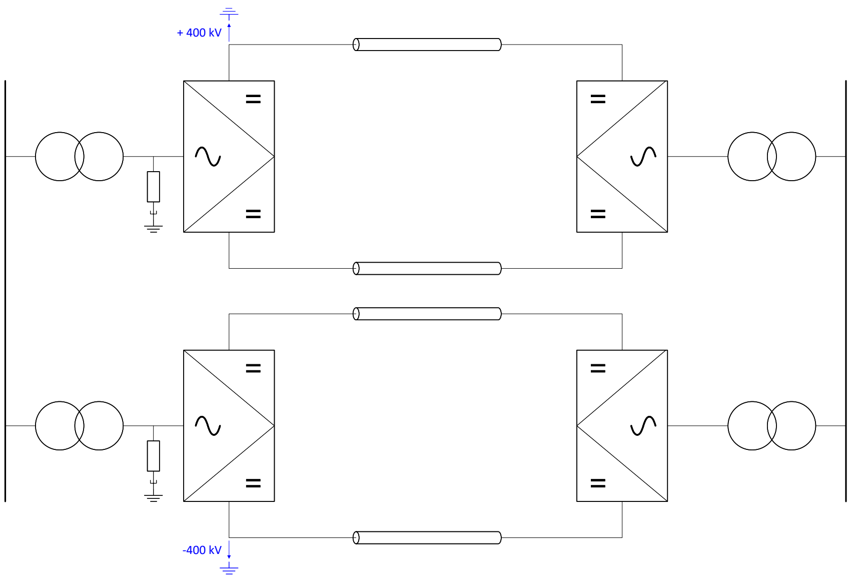

Figure 4: Dual symmetrical monopole (DSMP) configuration.

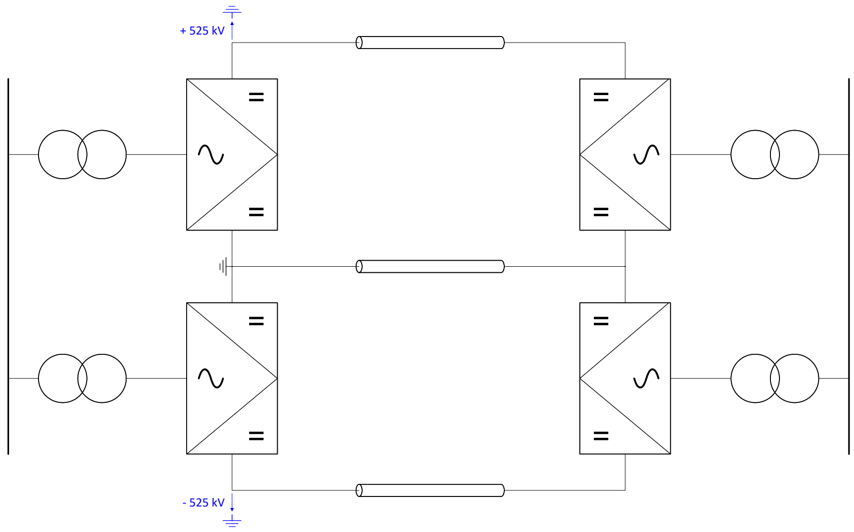

Figure 5: Bipole with dedicated metallic return (BIP) configuration.

2.4 Generation at rectifier

Organic Rankine Cycle (ORC) power generation systems will be used as the primary generation source. The individual generator units will be sized at app. 40 MW.

American Terawatt will provide a generator-system equivalent to be used for EMT-type studies and performance verification, together with an RSCAD model for hardware-in-the-loop (HIL) testing.

2.5 Load at inverter

The load connected to the HVDC system consists of data-center loads, which are characterized by power fluctuation. The system configuration will include compensation devices such as E-STATCOM or BESS which will compensate the power fluctuations.

American Terawatt will provide a data-center load equivalent to be used for EMT-type studies and performance verification, together with an RSCAD model for hardware-in-the-loop (HIL) testing.

3 AC data

The AC system data is flexible and is not fixed at the time of issue of this specification. A nominal AC voltage of 230 kV has been selected as a compromise between the insulation requirements and the current requirements. The supplier is encouraged to propose an alternative AC voltage where this results in a significant cost or performance benefit.

Although not connected to the AC grid, the interconnection requirements should follow the requirements defined in IEEE 2800 and PRC-029-2. The steady-state voltage levels are given in Table 2, temporary voltage range in Figure 6 and the transient voltage range in Table 3.

Concept verification studies are ongoing. The voltage and frequency range will be adapted to take into account the islanded network configuration.

	Unit	Value
Nominal system voltage	kV	230
Minimum continuous system voltage	kV	207 (0.9 p.u.)
Maximum continuous system voltage	kV	257 (1.1 p.u.)

Table 2: AC voltage

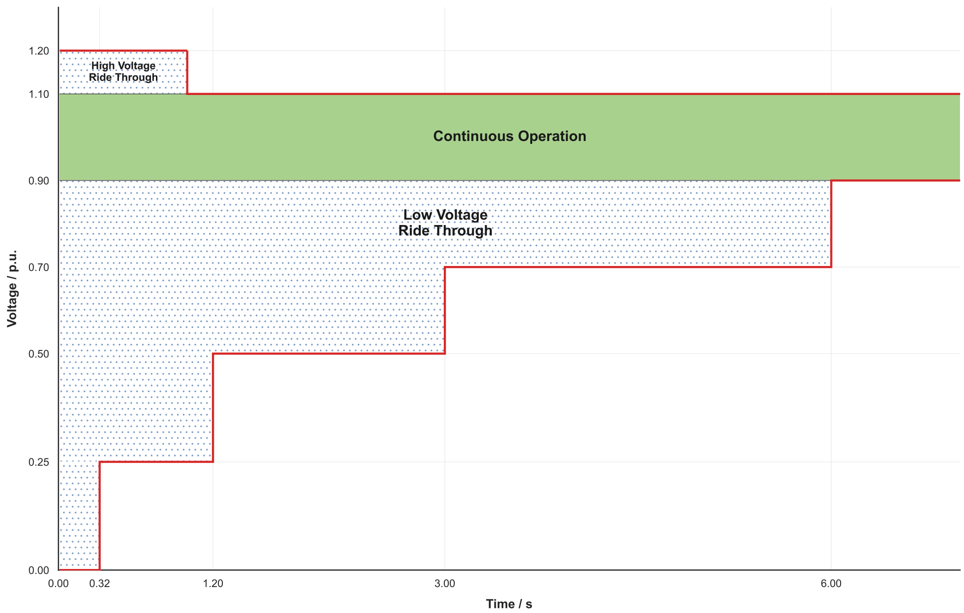

Figure 6: Temporary voltage range

	Unit	Value
Switching impulse withstand voltage	kV	850
Lightning impulse withstand voltage	kV	1050

Table 3: Transient voltages

The AC frequency range is given in Table 4. The continuous frequency range corresponds to the "no trip" range of PRC-029-2, within which the converter shall remain connected and in continuous operation at rated power without any restriction. Outside this range, the frequency ride-through capability shall follow the duration-dependent requirements of PRC-029-2. As noted above, the range will be reviewed once the concept verification studies for the islanded network configuration are completed.

	Unit	Value
Rated frequency	Hz	60
Minimum continuous frequency	Hz	58.8
Maximum continuous frequency	kV	61.2

Table 4: AC frequency

The short-circuit levels to be considered at the AC terminals of each station are given in Table 5. The values are subject to confirmation by the ongoing concept verification studies.

	Unit	Rectifier	Inverter
Minimum short-circuit level	MVA	SCR > 2	SCR < 1
Maximum short-circuit level	MVA	TBD	SCR = 1
Minimum X/R ratio	—	TBD	TBD
Maximum X/R ratio	—	TBD	TBD

Table 5: AC short-circuit levels

4 DC line data

The DC line route and the DC line design are under development at the time of issue of this specification; the assumptions below apply until the design is frozen.

It is assumed at this point in time that the majority of the route will be realized as overhead line (OHL). Depending on the outcome of the DC route evaluation, DC cable sections may be necessary if it allows the DC line permitting and execution. The resulting mixed OHL/cable configuration, together with the associated DC line parameters, will be provided by the end of the DC line design; for the corresponding dates refer to Section 7.

For the design of the generation and of the load it is assumed that clearance of a DC line fault is completed in $\leq 1.4\,s$. This value covers fault detection, converter blocking or current suppression, the de-energization of the faulted line, the deionization time and a successful restart of the link. The generation and the load are designed to ride through a disturbance of this duration; the HVDC system shall therefore be designed such that the fault clearance sequence, including any automatic restart attempt, does not exceed this time.

5 Performance requirements

5.1 Availability

5.1.1 Data-center availability requirement

The transmission system supplies data-center loads, whose availability is specified against the Uptime Institute tier framework. It is assumed that a **Tier III** requirement applies. Tier III describes a *concurrently maintainable* facility: every element of the power (and cooling) path can be maintained or replaced without interrupting supply to the load, achieved through redundant capacity components and multiple distribution paths (with one path active at a time). In availability terms, Tier III corresponds to about **99.982 %** availability — roughly **1.6 hours** of downtime per year.

A single HVDC link cannot, on its own, meet this target; achieving it requires redundancy at system level (e.g. dual full-capacity circuits) together with battery storage and, where necessary, on-site generation. The availability of the HVDC system is one contributor to the overall facility availability and is specified below.

5.1.2 HVDC terminals availability

For the overall availability performance, the assumed HVDC availability and performance values are given in Table 6. Operational performance shall be measured per CIGRE TB 956.

Metric	Description	Guaranteed
FEU	Forced energy unavailability	$\leq$0.5 %
SEU	Scheduled energy unavailability	$\leq$1.0 %
FOR	Forced outage rate SMP	$\leq$2 / yr
FOR	Forced outage rate BIP (single pole)	$\leq$2 / yr
FOR	Forced outage rate BIP (bipole)	$\leq$0.1 / yr
EA	Energy availability	$\geq$98.5 %

Table 6: HVDC availability and performance values.

6 Environmental and site conditions

The converter stations are located in Utah and Nevada. The environmental and site design conditions to be used for the design and rating of all equipment are given in Table 7.

Both sites are greenfield sites. No pre-existing structures, foundations, buried services or adjacent installations constrain the layout, and no restriction is placed on the station footprint, equipment arrangement. The supplier is therefore free to propose the station layout that gives the best overall techno-economic result. The final site boundaries, access roads and grading will be provided during the design phase; the supplier shall state the footprint and the maximum equipment weight to be brought on site.

Parameter	**Value**
Max. ambient temperature (T_{amb_max})	$+45$ ℃
Min. ambient temperature (T_{amb_min})	-35 ℃
Site elevation	1500 m
Design seismic level	IEEE 693-2018: Moderate

Table 7: Environmental and site design conditions.

7 Time Line

The following milestones apply to the project. Dates marked *[TBC]* are indicative and will be confirmed as the design develops. The milestones are listed in Table 8.

Ref.	Milestone	Date
M1	DC line environmental analisys and DC line design	Nov. 2026
M2	HVDC specification	Dec. 2026
M3	NTP Link 1	Jun. 2027
M4	NTP Link 2	Jun. 2027
M5	NTP Link 3	Jun. 2027
M6	NTP Link 4	Jun. 2027
M7	COD Link 1	Jun. 2030
M8	COD Link 2	Jun. 2030
M9	COD Link 3	Jun. 2030
M10	COD Link 4	Jun. 2030

Table 8: Project timeline milestones.